



CYBERSECURITY ROADMAP



STEP BY STEP GUIDE TO BECOME A CYBERSECURITY PROFESSIONAL

LEARN → PRACTICE → BUILD → SECURE

1		1. FOUNDATIONS - What is Cybersecurity? - Types of Cyber Attacks, Threats & Malware - CIA Triad (Confidentiality, Integrity, Availability) - Cybersecurity Career Paths	Duration 1-2 Weeks
2		2. NETWORKING BASICS - OSI Model, TCP/IP Model - IP Addressing, Subnetting - DNS, DHCP, HTTP/HTTPS - Firewalls, NAT, VPN Basics	Duration 2-3 Weeks
3		3. OPERATING SYSTEMS - Windows Fundamentals - Linux Fundamentals - Command Line (Bash/PowerShell) - File Systems, Permissions	Duration 2-3 Weeks
4		4. SECURITY FUNDAMENTALS - Authentication vs Authorization - Hashing, Encryption Basics - Public Key Infrastructure (PKI) - Security Policies & Best Practices	Duration 1-2 Weeks
5		5. THREAT & VULNERABILITY - Common Threats & Attack Vectors - Vulnerability Assessment Basics - CVE, CVSS Scoring - Threat Modeling	Duration 2-3 Weeks
6		6. ETHICAL HACKING BASICS - What is Ethical Hacking? - Reconnaissance Phases - Footprinting & Information Gathering - Scanning (Nmap Basics)	Duration 2-3 Weeks
7		7. SYSTEM HACKING - Enumeration - Exploitation Basics - Privilege Escalation - Maintaining Access	Duration 3-4 Weeks
8		8. WEB APPLICATION SECURITY - OWASP Top 10 - SQL Injection, XSS, CSRF - Burp Suite Basics - Web App Penetration Testing	Duration 3-4 Weeks
9		9. WIRELESS & MOBILE SECURITY - Wireless Network Basics - Wi-Fi Attacks (WEP, WPA, WPA2) - Mobile Security Basics - Bluetooth, NFC Security	Duration 2 Weeks
10		10. SECURITY OPERATIONS - SIEM Basics (Splunk / ELK) - Log Analysis - Incident Response Process - Digital Forensics Basics	Duration 2-3 Weeks
11		11. CLOUD SECURITY - Cloud Concepts (IaaS, PaaS, SaaS) - AWS / Azure / GCP Security Basics - IAM, Security Groups, Encryption - Cloud Security Best Practices	Duration 2-3 Weeks
12		12. ADVANCED TOPICS & CERTIFICATIONS - Penetration Testing Methodologies - Red Team vs Blue Team - Security Frameworks (NIST, ISO 27001) - Certifications (CompTIA Security+, CEH, OSCP, etc.)	Duration 3-4 Weeks

ESSENTIAL SKILLS

- Problem Solving
- Analytical Thinking
- Attention to Detail
- Continuous Learning
- Communication



IMPORTANT TOOLS



- Wireshark
- Nmap
- Metasploit
- Burp Suite
- John the Ripper
- Hashcat
- SQLMap
- Splunk / ELK

PRACTICE PLATFORMS



- TryHackMe
- Hack The Box
- PicoCTF
- OverTheWire
- PortSwigger Web Security Academy

LEARNING RESOURCES



- OWASP Documentation
- NIST Cybersecurity Framework
- MITRE ATT&CK
- The Cyber Mentor (YouTube)
- Practical Networking (YouTube)

CERTIFICATIONS ROADMAP



CAREER PATHS



- Security Analyst
- SOC Analyst
- Penetration Tester
- Security Engineer
- Incident Responder
- Security Consultant
- CISO



BUILD PROJECTS – PRACTICE LABS – CAPTURE THE FLAG (CTF)

Hands-on practice is the key to master cybersecurity!

